



GE Medical Systems

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Mobile LS 16 Service Intro Planner

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REVISION HISTORY

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Revision	Date	Reason for change
0	11/11/02	Initial Release

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SERVICE INTRODUCTION PLANNER (ISERV) LIGHTSPEED 16 MOBILE

Issuing Organization:

CT Engineering, GEMS-Am

1.0 OVERVIEW

This Introduction Planner provides information on the LightSpeed Ultra 4 8, and 16 slice products to be installed in non-fixed site environments.

This planner is intended to provide field service personnel with information regarding the LightSpeed ultra CT non-fixed site system going into forward production. This Service Introduction Planner is **NOT** a service manual, direction, or user guide. It is a collection of information taken from several source documents, ALL of which are available and should be used to install, maintain, and operate the CT LightSpeed Ultra non-fixed site system.

A **GE Medical System Engineering approved** non-fixed environment is any prefabricated, stand-alone environment, which is designed to hold a CT scanner. Non-fixed environments are shipped with the CT scanner preinstalled prior to the environment/scanner package being delivered to the customer site. Examples of non-fixed environments are mobile trailers, transportable trailers, relocatable buildings, and ISO containers.

The LightSpeed Ultra system for non-fixed environments is very similar to that for fixed site installations. Minor changes have been incorporated into the non-fixed version to accommodate the mobile environment needs (vibration, multi-hospital users on the same equipment, etc.)

1.1 PRODUCT STRUCTURE

Product Name	Engineering Program Name	Catalog Number	Description
Non-fixed LightSpeed Ultra 4-SLICE MOBILE	Helios 3 Mobile	B7854MB	Contains the following: B7854NG 4 slice GANTRY B7850NP PDU B7850NT H2 TABLE B7850MS SHIP COLLECTOR B7858MT H3 MOBILE SFT & TECH PUBS B7800KE ENGLISH KEYBOARD B7858CCH3 OCTANE II CONSOLE B7850LL LONG CABLE B7999PY UPS+HW KIT, MOBILE B7850MK LTSPD MOBILE KIT B7858LC LCD MONITOR 2PC/KIT B7530SP SMART PREP B7810WS VARISPEED B7810THLTSPD THIN TWIN HELICAL B7810TL LTSPD HELICAL TILT B7810RPLTSPD RECON PLUS K9050SF 8PT 10/100TX SWITCH K9000JR 10BASET PATCH CABLE KIT

Product Name	Engineering Program Name	Catalog Number	Description
			K9000KP 10M CAT V UTP PATCH CABLE B7858TE LTSPDUULTRA ENGL TECH PUBS
Non-fixed LightSpeed Ultra 8-SLICE MOBILE		B7858MB	Contains the following: B7858NG 4 slice GANTRY B7850NP PDU B7850NT H2 TABLE B7850MS SHIP COLLECTOR B7858MT H3 MOBILE SFT & TECH PUBS B7800KE ENGLISH KEYBOARD B7858CDH3 OCTANE II CONSOLE W/D3D B7850LL LONG CABLE B7999PY UPS+HW KIT, MOBILE B7850MK LTSPD MOBILE KIT B7858 8 SLICE SOFTWARE FOR OC B7810WS VARISPEED B7810TH LTSPD THIN TWIN HELICAL B7810TL LTSPD HELICAL TILT B7810RP LTSPD RECON PLUS B7858LC LCD MONITOR 2PC/KIT K9050SF 8PT 10/100TX SWITCH K9000JR 10BASET PATCH CABLE KIT K9000KP 10M CAT V UTP PATCH CABLE B7530SP SMART PREP B7810SS D3D SOFTWARE KIT FOR LS CONSOLE B7858TE LTSPDUULTRA ENGL TECH PUBS
Non-fixed LightSpeed Ultra 16-Slice Mobile		B7868MB	Contains the following: H2 TABLE ASM CATHAY PDU, LSPD PLUS LONG CABLE SET, LSPD PLUS ENGLISH KEYBOARD, SCIM H16 LS+ ENGLISH TECH PUBS LIGHTSPEED H16 GANTRY H16 CONSOLE GOC1 OCT2 D3D 16 SLICE OPTION SMART PREP (IRIX) VARISPEED, LSPD PLS (IRIX) 440MA OPTION (IRIX) LTSPD HELICAL TILT (IRIX) LTSPD RECON PLUS (IRIX) AUTO MA (IRIX) UPS+HW KIT, MOBILE LTSPD LCD MONITOR_2PC/KIT LTSPD MOBILE KIT 8PT 10/100TX SWITCH 10BASET PATCH CABLE KIT 10M CAT V UTP PATCH CABLE H16 MOBILE SHIP COLLECTOR H3 MOBILE SFT & TECH PUBS

1.2 SYSTEM SPECIFICATIONS

The Non-fixed LightSpeed Ultra configuration will have the same system performance specifications as the equivalent fixed site LightSpeed Ultra. All options specified in Non – Fixed CT LS system spec & Install requirements would be available for LightSpeed Ultra non-fixed site installations with the following exceptions:

1.3 KEY FEATURES OF NON-FIXED OFFERINGS

1.3.1 SERVICE FEATURES

The service features for the non-fixed version are virtually identical to the H2.1 fixed site features. There are no changes to the diagnostics or service procedures. Software changes are documented elsewhere within this ISERV.

1.3.2 CUSTOMER FEATURES

The customer features of the non-fixed LightSpeed Ultra systems are the same as for the fixed site equivalent. There are some additional features built into the customer software which will accommodate multi-hospital users on the same scanner, a modified shut down procedure which automatically tilts the gantry back for transport, the addition of flat panel monitors as standard monitors in non-fixed environments, and hardware to support the table for transport.

1.4 SERVICE AND OPERATOR DOCUMENTATION

*All service documents are available at the CT Service Engineering Web-site:
http://www.ct.med.ge.com/tech_pubs/welcome.htm*

Direction	Description	Classification
2341956-100	Pre-Installation/Site Planning Manual	General Class
2341957-100	Installation Manual	General Class
2341959-100	System Service Manual	General Class
2341961-100	Illustrated Parts Manual	General Class
2341962-100	Schematics	General Class
2341958-100	SW Load Procedure	General Class
2352974-100	Applications Tips and Work-Arounds	General Class
2351785-100	Technical Reference Manual	General Class
	Learning and Reference Guide	General Class
	Conformance statement for DICOM v3.0	
2341960-100	System Service Manual (Advanced), Rev 03	Advanced
2320440-100	Mobile Start Up /Shutdown	General Class
2315243-100	UPS Installation Mobile LightSpeed	General Class
2305587-100	Non-Fixed CT LS Systems Spec & Install Requirements	General Class
2305588-100	Non-Fixed CT LS Mechanical Integration Requirements	General Class
2341965-100	Non-Fixed CT LS 16 Electrical Integration Procedures	General Class
2318669-100	On-Site Mobile Set-up Procedure	General Class
2363105-100	Service Introduction Planner (Restricted)	Restricted

1.5 SUMMARY OF HARDWARE (MAIN DIFFERENCES VS. LS FIXED SITE)

- UPS and custom A1
- Flat Screen LCDs
- Net Work Switch
- No DASM (DICOM Print Only)
- Service Panel + Display for Gantry
- POGO Stick (table Transport)
- Modified Gantry Front cover Dollies

CUSTOMER-DRIVEN USER INTERFACE FEATURES

Flat panel monitors as standard

Transport warning switch: To trigger the transport lamp. Switch is physically mounted on STC side backstop

2.0 INTEGRATION

- Reconfig to enable mobile features
- Multi-Site set up
- Physical tilt limit: Can be found in Non – Fixed CT LS Electrical Integration Procedure 2318573-100 Rev01
- Mechanical and Electrical integration requirement for the non-fixed environment manufacturer can be found in the Non-Fixed CT LS Mechanical Integration requirement ref: 2305588-100 Rev01 and Non-Fixed CT LS Electrical Integration Procedures, 2318573-100 rev 01

The respective environment manufacturer provides site-specific pre-installation information. This information will include the pad requirements, power requirements and connector requirements, phone and network requirements, utility requirements (water, drains, etc. as required), and radiation safety zones as required at a minimum.

3.0 INSTALLATION

3.1 INSTALLATION TIME

Turnover time at customer delivery will be no longer than 6 hours at the customer site. There will be no further service involvement at subsequent customer locations. Power ground audit is required for all sites.

3.2 DOCUMENTS & TOOLS

Changes required to the LightSpeed Ultra Service documentation and Operators documentation will be made to the existing fixed site operators and service documentation. Separate documents will be developed for the non-fixed Mechanical, Electrical and installation requirements for non-fixed environment. The part numbers are as follow:

- 2305588-100 Non-Fixed CT LS Mechanical Integration Requirements
- 2318573-100 Non-Fixed CT LS Electrical Integration Procedure
- 2305587-100 Non-Fixed CT LS Systems Spec & Install Requirements

Gantry service tools are supplied in the install support kit. Mobile Gantry dollies may be different than fixed site dollies.

SUPPLIED BY ENVIRONMENT MANUFACTURER

F.E storage cabinet provided by the Van Company. The following tools will be stored in the Cabinet

- QA Phantom (Water Filled)
- 20cm Phantom
- 48cm Phantom
- Phantom Holder
- Gantry Display
- FE Box Purple **All Documentation**
- Rear Cover dollies
- Front Cover Dollies
- Install support kit (Box)
- Tube Hoist Kit
- Gantry Control Pad

REQUIRED TOOLS:

Items supplied by Field - standard tool kit as required to service the current LightSpeed Ultra fixed site systems.

Items supplied by the Environment Manufacturer - any special tools or dollies required to support the van and/or the LightSpeed Ultra system

Some van models may require a cabinet to be removed for major changes

3.3 ON-SITE SET UP PROCEDURE

On - Site set up procedure can be found in On-Site Mobile Set-up Procedure, Rev 00 Ref. 2318669-100

3.4 INSITE CONNECTION

InSite connection is mandatory on this system.

MODEM: Global modem to be shipped with each system

Standard weather resistance cables are supplied with each van

3.5 SOFTWARE

H3 M4 software will be employed in Mobile Ultra

4.0 SERVICE STRATEGY

4.1 SERVICE RESPONSIBILITIES

Reference 2318669-100 On-Site Mobile set up procedure

4.2 MAINTENANCE

Same as fixed-site

4.3 PREVENTIVE MAINTENANCE

Maintenance Frequency depends on Site activity.

Source pole recommendations are:

- In warranty: 4 PM visits/year
- Out of warranty: 1PM per 250 Kslices

Maintenance Time:

- In warranty: 13 hours/year.
- Out of warranty:

Required Tool: Items supplied by Field (Preliminary List)

Service Tool requirements will be the same as for the LightSpeed Ultra fixed site product. Any special tools (dollies, etc) that are required to support the scanner in a non-fixed environment will be supplied by the environment manufacturer.

PM Manual: **2141834**

4.4 CORRECTIVE MAINTENANCE

Correct maintenance strategy (tools, process, times, etc) for the non-fixed site version will be the same as for the fixed site LightSpeed Ultra.

Major Field Replaceable Units (FRU) are:

- X-Ray Tube
- High Voltage Tank
- Detector
- Slip Ring
- DAS

4.5 SPARE PARTS – ADDITIONS FOR MOBILE APPLICATION

Maintenance policy is stock at FRU level.

Part #	Description
Gantry	
2303589	Harness, tilt switch Assy.
2303601	Harness, ct mobile tilt
Table	
46-296317p4	OBC-STC & Table EMC power supply
2302366	Pogo Stick
Console	
2306872	18.1" Global Standard LCD Monitor
PDU	
2310005	HARDENED UPS FOR MOBILE LIGHTSPEED +

5.0 TRAINING

There will be no specific training class for the LightSpeed Ultra in non-fixed site installations. There will be a T-CON with primary F.E to discuss the differences between the Fixed site and Non-Fixed site services.

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